

# C S Vaishakh

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## PROFESSIONAL SUMMARY

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AI & Data Science undergraduate student with a strong interest in backend development, systems architecture, and machine learning. Experienced in building RESTful APIs with Go (Fiber) and Node.js (Express), managing relational databases like PostgreSQL, and developing full-stack web applications.

## EDUCATION

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**St. Joseph's College of Engineering and Technology, Palai**

Kottayam, Kerala

*B.Tech in Artificial Intelligence and Data Science — APJ Abdul Kalam Technological University*

2023 – 2027

## PROJECTS

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### Quickhand – Handyman Hiring Platform

December 2025 – Present

- Implemented role-based access control (RBAC) across two user roles, securing Fiber API routes and verifying sessions with stateless JWTs.
- Engineered a high-performance monorepo using a Go backend (Fiber v3) and a React frontend (TanStack Start) orchestrated via Bun.
- Technologies:** Go, Fiber v3, React, TanStack Start, TypeScript, Tailwind CSS, PostgreSQL, GORM, Docker

### Course Selling Platform – Backend System

November 2025

- Developed a RESTful backend API in Node.js and Express.js (v5) supporting course lifecycle operations across admin and student roles.
- Designed database schemas and relationships using MongoDB/Mongoose to track users, courses, and course enrollments.
- Implemented stateless JWT-based authentication and Bcrypt password hashing to secure endpoints and restrict role actions.
- Technologies:** Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt

### ONT Monitor – ONT Connection Tracker

April 2026

- Built a Go-based microservice using Fiber v3 and GORM to ingest and process real-time status change events for ONTs.
- Implemented thread-safe in-memory queues and a background worker pool to evaluate outages and prevent race conditions.
- Integrated Twilio API to dispatch automated outage and recovery WhatsApp alerts to registered customers upon state transitions.
- Technologies:** Go, Fiber v3, GORM, PostgreSQL, Twilio API, Docker

### PortfolioIQ – Portfolio Rebalancing Platform

December 2025 – April 2026

- Engineered a privacy-preserving portfolio rebalancing platform structured as a monorepo using npm workspaces.
- Designed a client-side training pipeline using TensorFlow.js to train local logistic regression models on a 12-dimensional feature vector, eliminating raw data uploads.
- Implemented the Federated Averaging (FedAvg) algorithm on the Express.js aggregator to synchronize local updates with a FastAPI scikit-learn model.
- Technologies:** Next.js, Express.js, FastAPI, TensorFlow.js, Drizzle ORM, PostgreSQL, Python, TypeScript, Turborepo, Docker

## TECHNICAL SKILLS

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**Programming:** Python, TypeScript, Go, Java, JavaScript

**Web & Backend:** Next.js, React, Express.js, Fiber v3, FastAPI, Node.js, TanStack Start, REST APIs

**Databases & ORMs:** PostgreSQL, MongoDB, GORM, Drizzle ORM, Mongoose

**AI / ML:** TensorFlow.js, Feature Engineering, Model Evaluation, scikit-learn, NumPy, Pandas

**Tools:** Git, GitHub Actions, Docker, Turborepo, Bun, Postman, Linux, VS Code